

Smart Garage Gate — MicroPython Code

```
1 from machine import Pin
2 from time import sleep_ms
3
4 ENTRY_BUTTON = 14
5 EXIT_BUTTON = 27
6
7 YELLOW_LED = 5
8 GREEN_LED = 2
9 BLUE_LED = 18
10 RED_LED = 4
11
12 MAX_CAPACITY = 15
13
14 entry_button = Pin(ENTRY_BUTTON, Pin.IN, Pin.PULL_UP)
15 exit_button = Pin(EXIT_BUTTON, Pin.IN, Pin.PULL_UP)
16
17 yellow_led = Pin(YELLOW_LED, Pin.OUT)
18 green_led = Pin(GREEN_LED, Pin.OUT)
19 blue_led = Pin(BLUE_LED, Pin.OUT)
20 red_led = Pin(RED_LED, Pin.OUT)
21
22 car_count = 0
23
24
25 def update_status():
26     green_led.value(car_count < MAX_CAPACITY)
27     red_led.value(car_count >= MAX_CAPACITY)
28
29
30 def blink(led):
31     led.on()
32     sleep_ms(250)
33     led.off()
34
35
36 def show_status():
37     print()
38     print("=====")
39     print("      FUTURE MALL GARAGE")
40     print("=====")
41     print(" Cars      : {}/{}".format(car_count, MAX_CAPACITY))
42     print(" Available : {}".format(MAX_CAPACITY - car_count))
43
44     if car_count >= MAX_CAPACITY:
45         print(" Status      : FULL")
46     else:
47         print(" Status      : AVAILABLE")
48
49     print("=====")
50
51
52 update_status()
53 show_status()
54
55 while True:
56
57     if entry_button.value() == 0:
58
59         if car_count < MAX_CAPACITY:
60             car_count += 1
61             blink(yellow_led)
62             update_status()
63
64             print(">> ENTRY")
65             show_status()
66
67         while entry_button.value() == 0:
68             sleep_ms(20)
69
70     if exit_button.value() == 0:
71
72         if car_count > 0:
73             car_count -= 1
74             blink(blue_led)
75             update_status()
76
77             print(">> EXIT")
78             show_status()
79
80         while exit_button.value() == 0:
81             sleep_ms(20)
82
83     sleep_ms(20)
```

Smart Garage Gate — Wiring & Operation

